SentinelVision AI

Volume 05

REST API Documentation

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title: REST API Documentation

status: engineering-ready

version: 2.0.0

> **SentinelVision AI — Volume 5**

> Status: Implemented | Planned | Future

1. Document Control

Field	Value
Document ID	DOC-API-005
Version	2.0.0
Status	Engineering Ready
Base URL (local)	`http://localhost:8000`
API Prefix	`/api/v1/`
OpenAPI Schema	`GET /api/schema/`
Swagger UI	`GET /api/docs/`

2. API Conventions

Convention	Specification
Protocol	HTTP/1.1 (TLS 1.2+ in production)
Content-Type	`application/json`
Charset	UTF-8
Timestamps	ISO 8601 UTC (`2026-07-17T07:00:00.000000Z`)
Primary keys	UUID v4 in path and response
Pagination	PageNumberPagination, `PAGE_SIZE=25` (Implemented); cursor Planned
Filtering	`django-filter` query params where documented
Ordering	`?ordering=field` or `?ordering=-field`
Search	`?search=term` where documented

2.1 Authentication Methods

Method	Header	Used By	Status
JWT Bearer	`Authorization: Bearer <access_token>`	Dashboard users	Implemented

Method	Header	Used By	Status
Edge API Key	`X-Edge-API-Key: <raw_key>`	Edge workers	Implemented
Edge Alt	`Authorization: Edge <raw_key>`	Edge workers	Implemented

2.2 JWT Token Policy

Token	Lifetime	Notes
Access	15 minutes	Used for API calls
Refresh	7 days	`ROTATE_REFRESH_TOKENS=True`; blacklist Planned

2.3 Role Permissions Matrix

Capability	Owner	Admin	Operator	Viewer
Read org/sites/cameras/events/alerts	Yes	Yes	Yes	Yes
Create/update/delete sites, zones, cameras	Yes	Yes	No	No
Create edge devices	Yes	Yes	No	No
Create alert rules	Yes	Yes	No	No
Review events	Yes	Yes	Yes	No
Acknowledge/resolve alerts	Yes	Yes	Yes	No
List audit logs	Yes	Yes	Yes	Yes

2.4 Standard Error Responses

Current (DRF default) for most endpoints:

```
{
"detail": "Authentication credentials were not provided."
}
```

```
{
"field_name": ["This field is required."]
}
```

Target envelope (Planned — use for new endpoints):

```
{
"error": {
"code": "VALIDATION_ERROR",
"message": "Human-readable summary",
"details": {"field": ["reason"]}
}
}
```

HTTP	Code (Planned)	Meaning
400	VALIDATION_ERROR	Invalid request body/query
401	UNAUTHENTICATED	Missing or invalid credentials

HTTP	Code (Planned)	Meaning
403	FORBIDDEN	Authenticated but insufficient role/tenant
404	NOT_FOUND	Resource does not exist or not visible
409	CONFLICT	Duplicate / state conflict
429	RATE_LIMITED	Too many requests (Planned)
500	INTERNAL_ERROR	Unexpected server error

2.5 Pagination Envelope

```
{
"count": 100,
"next": "http://localhost:8000/api/v1/events/?page=2",
"previous": null,
"results": []
}
```

3. Health & Schema

API-HEALTH — Health Check

Field	Value
Method / URL	`GET /api/v1/health/`
Auth	None
Permissions	Public
Status	Implemented

Response `200`**:**

```
{
"status": "healthy",
"service": "sentinelvision-api",
"timestamp": "2026-07-17T07:00:00.123456+00:00"
}
```

API-SCHEMA — OpenAPI Schema

Field	Value
Method / URL	`GET /api/schema/`
Auth	None
Format	OpenAPI 3.0 YAML/JSON
Status	Implemented via `drf-spectacular`

API-DOCS — Swagger UI

Field	Value
Method / URL	`GET /api/docs/`
Auth	None
Status	Implemented

4. Authentication Endpoints

API-AUTH-TOKEN — Obtain JWT Pair

Field	Value
Method / URL	`POST /api/v1/auth/token/`
Auth	None
Status	Implemented

Request:

```
{
"username": "admin",
"password": "admin123"
}
```

Validation:

Field	Rules
username	Required, string
password	Required, string

Response `200`:

```
{
"access": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
"refresh": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..."
}
```

Errors:

Status	Condition
400	Missing fields
401	Invalid credentials

API-AUTH-REFRESH — Refresh Access Token

Field	Value
Method / URL	`POST /api/v1/auth/token/refresh/`
Auth	None (body carries refresh)
Status	Implemented

Request:

```
{
```

```
"refresh": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..."
}
```

Response `200`:

```
{
"access": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
"refresh": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..."
}
```

5. Tenancy Endpoints

API-ORG-LIST — List Organizations

Field	Value
Method / URL	`GET /api/v1/organizations/`
Auth	JWT
Permissions	Authenticated; scoped to memberships
Status	Implemented

Response `200` (paginated):

```
{
"count": 1,
"next": null,
"previous": null,
"results": [
{
"id": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
"name": "ACME Security Corp",
"slug": "acme-security",
"settings": {"retention_days": 90},
"is_active": true,
"created_at": "2026-07-17T06:00:00Z",
"updated_at": "2026-07-17T06:00:00Z"
}
]
}
```

API-MEMBERSHIP-ME — Current User Memberships

Field	Value
Method / URL	`GET /api/v1/memberships/me/`
Auth	JWT
Status	Implemented

Response `200`:

```
[
{
"organization": {
"id": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
"name": "ACME Security Corp",
"slug": "acme-security",
"settings": {"retention_days": 90},
"is_active": true,
"created_at": "2026-07-17T06:00:00Z",
"updated_at": "2026-07-17T06:00:00Z"
},
"role": "owner"
```

```
}
]
```

Role enum: `owner` | `admin` | `operator` | `viewer`

6. Dashboard

API-DASH-SUMMARY — Overview Metrics

Field	Value
Method / URL	`GET /api/v1/dashboard/summary/`
Auth	JWT
Status	Implemented

Response `200`:

```
{
"cameras_total": 1,
"cameras_online": 1,
"events_today": 12,
"events_pending_review": 5,
"alerts_open": 2,
"alerts_critical": 1,
"edge_devices_online": 1,
"edge_devices_total": 1
}
```

Scoping: Aggregates only resources in organizations where the user has membership.

7. Sites

API-SITE-LIST — List / Create Sites

Field	Value
Method / URL	`GET /api/v1/sites/` / `POST /api/v1/sites/`
Auth	JWT
GET permissions	Member of org
POST permissions	Owner or Admin of target organization
Status	Implemented

Query params (GET):

Param	Description
search	Match `name`, `address`
ordering	`name`, `created_at`

POST request:

```
{
"organization": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
"name": "Warehouse B",
"address": "200 Industrial Rd",
"timezone": "America/New_York",
```

```
"is_active": true
}
```

Validation:

Field	Rules
organization	Required UUID; must exist; user must be Owner/Admin
name	Required; unique per organization
address	Optional string
timezone	Optional; default UTC
is_active	Optional boolean; default true

Response `201`:

```
{
"id": "11111111-2222-3333-4444-555555555555",
"organization": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
"organization_name": "ACME Security Corp",
"name": "Warehouse B",
"address": "200 Industrial Rd",
"timezone": "America/New_York",
"is_active": true,
"created_at": "2026-07-17T07:10:00Z",
"updated_at": "2026-07-17T07:10:00Z"
}
```

Errors: `400` validation; `403` insufficient role; `401` unauthenticated.

API-SITE-DETAIL — Retrieve / Update / Delete Site

Field	Value		
Method / URL	`GET	PATCH	DELETE /api/v1/sites/{id}/`
Auth	JWT		
PATCH/DELETE	Owner or Admin		
Status	Implemented		

PATCH example:

```
{
"address": "200 Industrial Rd, Suite 4",
"is_active": false
}
```

8. Zones

API-ZONE-LIST — List / Create Zones

Field	Value
Method / URL	`GET /api/v1/zones/` / `POST /api/v1/zones/`
Auth	JWT
POST	Owner or Admin of site's organization
Status	Implemented

Query params: `site`, `is_restricted`, `is_active`

POST request:

```
{
"site": "11111111-2222-3333-4444-555555555555",
"name": "Loading Dock",
"polygon_json": [[0.1, 0.2], [0.8, 0.2], [0.8, 0.9], [0.1, 0.9]],
"is_restricted": true,
"is_active": true
}
```

Validation:

Field	Rules
site	Required; must belong to user's org
name	Required; unique per site
polygon_json	Array of `[x,y]` pairs; preferably normalized 0–1
is_restricted	Boolean
is_active	Boolean

Response `201`**:** includes `id`, `site`, `site_name`, `organization`, zone fields, timestamps.

API-ZONE-DETAIL — Retrieve / Update / Delete Zone

Field	Value		
Method / URL	`GET	PATCH	DELETE /api/v1/zones/{id}/`
PATCH/DELETE	Owner or Admin		
Status	Implemented		

9. Cameras

API-CAM-LIST — List / Create Cameras

Field	Value
Method / URL	`GET /api/v1/cameras/` / `POST /api/v1/cameras/`
Auth	JWT
POST	Owner or Admin
Status	Implemented

Query params: `site`, `status`, `is_active`, `search` (name)

POST request:

```
{
"site": "11111111-2222-3333-4444-555555555555",
"zone": "aaaaaaaa-bbbb-cccc-dddd-eeeeeeeeeeee",
"name": "Dock Camera 02",
"rtsp_url": "rtsp://user:pass@192.168.1.50:554/stream1",
"enabled_modules": ["multi_person_tracking", "intrusion_detection"],
"metadata": {"vendor": "Hikvision", "resolution": "1080p"},
"is_active": true
}
```

Validation:

Field	Rules
site	Required UUID
zone	Optional; **should** belong to same site (app validation Planned)
name	Required; unique per site
rtsp_url	Optional string (max 512)
enabled_modules	JSON array of module keys
status / last_seen_at	Read-only on create

Module keys: `face_recognition`, `multi_person_tracking`, `activity_recognition`, `intrusion_detection`, `unknown_person`, `visitor_management`, `attendance`, `fire_smoke`, `weapon_detection`

Response `201`**:**


```
{
"id": "cccccccc-dddd-eeee-ffff-000000000001",
"site": "11111111-2222-3333-4444-555555555555",
"site_name": "Warehouse B",
"organization": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
"zone": "aaaaaaaa-bbbb-cccc-dddd-eeeeeeeeeeee",
"zone_name": "Loading Dock",
"name": "Dock Camera 02",
"rtsp_url": "rtsp://user:pass@192.168.1.50:554/stream1",
"status": "offline",
"enabled_modules": ["multi_person_tracking", "intrusion_detection"],
"last_seen_at": null,
"metadata": {"vendor": "Hikvision", "resolution": "1080p"},
"is_active": true,
"created_at": "2026-07-17T07:20:00Z",
"updated_at": "2026-07-17T07:20:00Z"
}
```


API-CAM-DETAIL — Retrieve / Update / Delete Camera

Field	Value		
Method / URL	`GET	PATCH	DELETE /api/v1/cameras/{id}/`
PATCH/DELETE	Owner or Admin		
Status	Implemented		

10. Edge Devices

API-EDGE-DEV-LIST — List / Register Edge Device

Field	Value
Method / URL	`GET /api/v1/edge-devices/` / `POST /api/v1/edge-devices/`
Auth	JWT
POST	Owner or Admin

Field	Value
Status	Implemented

POST request:

```
{
"site": "11111111-2222-3333-4444-555555555555",
"name": "Edge Node WH-B-01",
"metadata": {"gpu": "RTX 3060"},
"is_active": true
}
```

Response `201` (api_key shown once):

```
{
"id": "edge0001-0000-0000-0000-000000000001",
"site": "11111111-2222-3333-4444-555555555555",
"site_name": "Warehouse B",
"organization": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
"name": "Edge Node WH-B-01",
"status": "offline",
"last_seen_at": null,
"version": "",
"metadata": {"gpu": "RTX 3060"},
"is_active": true,
"api_key_prefix": "AbCdEfGh",
"api_key": "AbCdEfGhIjKlMnOpQrStUvWxYz0123456789_full",
"created_at": "2026-07-17T07:25:00Z",
"updated_at": "2026-07-17T07:25:00Z"
}
```

Security notes:

- Store `api_key` only on the edge node; never log it again.
- DB stores `api_key_hash` (SHA-256), not plaintext.
- Key rotation: Planned.

API-EDGE-HEARTBEAT — Edge Heartbeat

Field	Value
Method / URL	`POST /api/v1/edge/heartbeat/`
Auth	Edge API Key
Status	Implemented

Headers:

```
X-Edge-API-Key: <raw_api_key>
Content-Type: application/json
```

Request:

```
{
"version": "0.1.0",
"status": "online",
"metadata": {"service": "edge-ai", "mode": "simulation"}
}
```

Response `200`:

```
{
"status": "ok",
"registered": true,
"device_id": "edge0001-0000-0000-0000-000000000001"
}
```

If key maps to no device: `{"status":"ok","registered":false}`.

Side effects: Updates `EdgeDevice.last_seen_at`, `status`, `version`; marks site cameras online.

API-EDGE-CAM-LIST — Cameras for Edge Site

Field	Value
Method / URL	`GET /api/v1/edge/cameras/`
Auth	Edge API Key
Status	Implemented

Response `200`:

```
{
"results": [
{
"id": "cccccccc-dddd-eeee-ffff-000000000001",
"name": "Dock Camera 02",
"rtsp_url": "rtsp://...",
"status": "online",
"enabled_modules": ["multi_person_tracking"]
}
]
}
```

Empty `results` if edge key is not bound to a device.

11. Detection Events

API-EVT-LIST — List Events

Field	Value
Method / URL	`GET /api/v1/events/`
Auth	JWT
Status	Implemented

Query params:

Param	Description			
camera	UUID filter			
event_type	Enum filter			
severity	`info` |	`warning` |	`critical`	
review_status	`pending` |	`confirmed` |	`false_positive` |	`escalated`
search	camera name or edge_event_id			
ordering	`created_at`, `confidence`, `severity` (prefix `-` for desc)			
page	Page number			

Response item:

```
{
"id": "evt00001-0000-0000-0000-000000000001",
"camera": "cccccccc-dddd-eeee-ffff-000000000001",
"camera_name": "Dock Camera 02",
```

```
"site_name": "Warehouse B",
"organization": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
"zone": "aaaaaaaa-bbbb-cccc-dddd-eeeeeeeeeeee",
"event_type": "person_detected",
"severity": "warning",
"confidence": 0.87,
"model_name": "yolo-sim",
"model_version": "0.1.0",
"payload": {"track_id": 1, "bbox": [100, 100, 200, 300]},
"review_status": "pending",
"reviewed_by": null,
"reviewed_at": null,
"edge_event_id": "edge-evt-abc-123",
"created_at": "2026-07-17T07:30:00Z",
"updated_at": "2026-07-17T07:30:00Z"
}
```


Event type enum:

`person_detected` | `face_recognized` | `unknown_person` | `intrusion` | `activity` | `fire_smoke` | `weapon` | `attendance` | `visitor`

API-EVT-INGEST — Ingest Event (Edge)

Field	Value
Method / URL	`POST /api/v1/events/ingest/`
Auth	Edge API Key
Status	Implemented

Request:


```
{
"camera_id": "cccccccc-dddd-eeee-ffff-000000000001",
"event_type": "person_detected",
"severity": "warning",
"confidence": 0.87,
"model_name": "yolo-sim",
"model_version": "0.1.0",
"payload": {
"track_id": 1,
"bbox": [100, 100, 200, 300],
"simulated": true
},
"zone_id": "aaaaaaaa-bbbb-cccc-dddd-eeeeeeeeeeee",
"edge_event_id": "edge-evt-abc-123"
}
```


Validation:

Field	Rules
camera_id	Required UUID; active camera
event_type	Required enum
severity	Optional enum; default `info`
confidence	Float 0.0–1.0
model_name	Optional string ≤128
model_version	Optional string ≤64
payload	Optional object
zone_id	Optional UUID
edge_event_id	Optional; used for idempotency

Business rules:

1. Camera must exist and `is_active=true`.
2. If edge user has bound device, camera.site must equal device.site → else `403`.
3. If `edge_event_id` matches existing → return `200` with existing event (idempotent).
4. On create → evaluate alert rules; may create `Alert` rows.
5. Updates camera `last_seen_at` and `status=online`.

Responses:

Status	Meaning
201	New event created
200	Idempotent hit
403	Camera not on edge site
404	Camera not found
400	Validation failure

Remediation Planned: Scope idempotency by organization; reject zone not belonging to camera.site; remove unbound global `EDGE_API_KEY` fallback for ingest.

API-EVT-REVIEW — Review Event

Field	Value	
Method / URL	`GET	PATCH /api/v1/events/{id}/review/`
Auth	JWT	
PATCH	Owner, Admin, or Operator	
Status	Implemented	

PATCH request:

```
{
"review_status": "confirmed"
}
```

Allowed values: `pending` | `confirmed` | `false_positive` | `escalated`

Side effects: Sets `reviewed_by` = current user, `reviewed_at` = now.

12. Alerts

API-ALERT-RULE-LIST — List / Create Alert Rules

Field	Value
Method / URL	`GET /api/v1/alert-rules/` / `POST /api/v1/alert-rules/`
Auth	JWT
POST	Owner or Admin
Status	Implemented

Query params: `event_type`, `is_active`

POST request:

```
{
"organization": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
"name": "Dock Intrusion Critical",
"event_type": "intrusion",
"min_severity": "critical",
"min_confidence": 0.7,
"is_active": true,
"notify_email": true,
"conditions": {"sites": ["11111111-2222-3333-4444-555555555555"]}
}
```

Alert evaluation (Implemented): For each active rule matching `event_type`, if event severity rank ≥ rule min_severity and confidence ≥ min_confidence → create Alert.

Note: `notify_email` and `conditions` are stored but not yet dispatched (notification service Planned).

API-ALERT-LIST — List Alerts

Field	Value
Method / URL	`GET /api/v1/alerts/`
Auth	JWT
Status	Implemented

Query params: `status`, `severity`

Response item:

```
{
"id": "alert001-0000-0000-0000-000000000001",
"organization": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
"event": "evt00001-0000-0000-0000-000000000001",
"event_type": "person_detected",
"camera_name": "Dock Camera 02",
"site_name": "Warehouse B",
"rule": "rule0001-0000-0000-0000-000000000001",
"title": "Person Detected — Dock Camera 02",
"message": "Detection at Warehouse B with confidence 0.87",
"severity": "warning",
"status": "open",
"acknowledged_by": null,
"acknowledged_at": null,
"resolved_at": null,
"created_at": "2026-07-17T07:30:01Z",
"updated_at": "2026-07-17T07:30:01Z"
}
```

API-ALERT-ACK — Acknowledge / Resolve Alert

Field	Value
Method / URL	`PATCH /api/v1/alerts/{id}/acknowledge/`
Auth	JWT
Permissions	Owner, Admin, Operator
Status	Implemented

Request:

```
{
"status": "acknowledged"
}
```

Allowed status values: `open` | `acknowledged` | `resolved` | `dismissed`

Side effects:

- `acknowledged` → set `acknowledged_by`, `acknowledged_at`
- `resolved` or `dismissed` → set `resolved_at`

13. Audit Logs

API-AUDIT-LIST — List Audit Logs

Field	Value
Method / URL	`GET /api/v1/audit-logs/`
Auth	JWT
Status	Partial — endpoint Implemented; auto-write Planned

Query params: `action`, `resource_type`

Response item:

```
{
"id": "aud00001-0000-0000-0000-000000000001",
"organization": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
"actor": 1,
"actor_username": "admin",
"action": "camera.create",
"resource_type": "camera",
"resource_id": "cccccccc-dddd-eeee-ffff-000000000001",
"ip_address": "127.0.0.1",
"details": {},
"created_at": "2026-07-17T07:20:00Z"
}
```

14. Planned Endpoints (M3–M6)

API-ATT-LIST — Attendance Records

`GET /api/v1/attendance/?person_id=&date_from=&date_to=` — JWT, HR+/Admin — Planned

API-VIS-LIST — Visitors

`GET|POST /api/v1/visitors/` — JWT, HR+/Admin — Planned

API-PERSON-ENROLL — Face Enrollment

`POST /api/v1/persons/{id}/enroll/` — JWT, Admin — multipart image + consent_id — Planned

API-RPT-EXPORT — Report Export

`POST /api/v1/reports/export/` — JWT, Admin+ — `{type, date_from, date_to, format}` — Planned

API-EVIDENCE-GET — Evidence Artifact

`GET /api/v1/evidence/{id}/` — JWT, Operator+ — signed URL to object storage — Planned

15. OpenAPI Generation

```
cd backend
python manage.py spectacular --file ../docs/openapi-schema.yml
```

Swagger UI: `http://localhost:8000/api/docs/`

CI requirement: Schema generation must succeed without import errors.

15.1 Example OpenAPI Path Fragment

```
/api/v1/events/ingest/:
post:
operationId: events_ingest_create
tags: [events]
security:
- EdgeApiKey: []
requestBody:
required: true
content:
application/json:
schema:
$ref: '#/components/schemas/EventIngest'
responses:
'201':
description: Created
'200':
description: Idempotent existing event
```

16. Client Integration Notes

Client	Base	Auth
React dashboard	`/api/v1` (proxied)	JWT access in `Authorization`
Edge service	`http://backend:8000/api/v1`	`X-Edge-API-Key`
External integrations	Public HTTPS base	JWT service account Planned

Frontend token storage (current): access token in `localStorage` key `sv_access_token`. Refresh retry Planned.

17. Rate Limits (Planned)

Client	Limit
User JWT	100 req/min
Edge device	1000 req/min
Ingest burst	50 events/sec per edge

18. Remediation Checklist for API Hardening

ID	Issue	Target Fix
API-FIX-01	Global `edge_event_id` idempotency	Unique per organization
API-FIX-02	Unbound EDGE_API_KEY ingest	Require device-bound key
API-FIX-03	Zone/site mismatch on camera/event	Serializer validation

ID	Issue	Target Fix
API-FIX-04	Inconsistent error envelope	Adopt standard error schema
API-FIX-05	Audit not written on mutations	Call `log_audit` in views
API-FIX-06	No rate limiting	Add DRF throttling classes

19. Traceability

API ID	Requirements	Use Cases	Tests
API-AUTH-TOKEN	UC-AUTH-01	—	TEST-API-002
API-CAM-LIST	FR-CAM-01	UC-CAM-01	TEST-API-003
API-EVT-INGEST	FR-ALERT-01	UC-EVT-01	TEST-API-004,005
API-EVT-REVIEW	US-02	UC-EVT-02	Planned
API-ALERT-ACK	FR-ALERT-04	UC-ALERT-01	Planned E2E
API-DASH-SUMMARY	FR-DASH-01	—	TEST-API-006